

Auteur: Ahmed Meiyaki
Studierichting: Informatica
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$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{\tan 5x}{\sin 2x} \\ &= \lim_{x \rightarrow 0} \frac{x \cdot \tan 5x}{x \cdot \sin 2x} \\ &= \lim_{x \rightarrow 0} \frac{\tan 5x}{x} \cdot \frac{x}{\sin 2x} \\ &= \lim_{x \rightarrow 0} 5 \left(\frac{\tan 5x}{5x} \right) \cdot \lim_{x \rightarrow 0} \left(2 \left(\frac{\sin 2x}{2x} \right) \right)^{-1} = \frac{5}{2} \end{aligned}$$

References